Test Series Code

06

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 22/02/2026

ANSWER KEY

1.	(1)	46.	(1)	91.	(1)	136.	(1)
2.	(2)	47.	(4)	92.	(2)	137.	(3)
3.	(3)	48.	(2)	93.	(2)	138.	(3)
4.	(2)	49.	(2)	94.	(4)	139.	(3)
5.	(2)	50.	(4)	95.	(3)	140.	(2)
6.	(1)	51.	(2)	96.	(4)	141.	(2)
7.	(4)	52.	(2)	97.	(1)	142.	(3)
8.	(3)	53.	(2)	98.	(2)	143.	(2)
9.	(2)	54.	(3)	99.	(1)	144.	(2)
10.	(2)	55.	(4)	100.	(3)	145.	(1)
11.	(3)	56.	(3)	101.	(1)	146.	(2)
12.	(1)	57.	(3)	102.	(2)	147.	(4)
13.	(1)	58.	(3)	103.	(2)	148.	(4)
14.	(1)	59.	(1)	104.	(2)	149.	(2)
15.	(3)	60.	(2)	105.	(1)	150.	(3)
16.	(1)	61.	(2)	106.	(4)	151.	(3)
17.	(3)	62.	(1)	107.	(3)	152.	(3)
18.	(1)	63.	(3)	108.	(1)	153.	(2)
19.	(2)	64.	(1)	109.	(3)	154.	(2)
20.	(3)	65.	(3)	110.	(2)	155.	(2)
21.	(2)	66.	(2)	111.	(2)	156.	(2)
22.	(1)	67.	(4)	112.	(4)	157.	(1)
23.	(3)	68.	(1)	113.	(1)	158.	(2)
24.	(4)	69.	(2)	114.	(3)	159.	(2)
25.	(3)	70.	(2)	115.	(1)	160.	(3)
26.	(3)	71.	(2)	116.	(4)	161.	(2)
27.	(3)	72.	(2)	117.	(4)	162.	(1)
28.	(3)	73.	(1)	118.	(2)	163.	(3)
29.	(2)	74.	(2)	119.	(2)	164.	(1)
30.	(2)	75.	(2)	120.	(1)	165.	(2)
31.	(3)	76.	(1)	121.	(1)	166.	(4)
32.	(4)	77.	(1)	122.	(3)	167.	(2)
33.	(1)	78.	(2)	123.	(4)	168.	(2)
34.	(2)	79.	(3)	124.	(3)	169.	(2)
35.	(3)	80.	(2)	125.	(4)	170.	(3)
36.	(4)	81.	(2)	126.	(1)	171.	(1)
37.	(2)	82.	(3)	127.	(3)	172.	(3)
38.	(2)	83.	(2)	128.	(1)	173.	(4)
39.	(3)	84.	(3)	129.	(2)	174.	(2)
40.	(4)	85.	(2)	130.	(4)	175.	(1)
41.	(1)	86.	(1)	131.	(2)	176.	(2)
42.	(1)	87.	(3)	132.	(4)	177.	(1)
43.	(3)	88.	(2)	133.	(1)	178.	(4)
44.	(2)	89.	(2)	134.	(3)	179.	(3)
45.	(1)	90.	(2)	135.	(2)	180.	(2)

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Hints & Solutions

1. (1)

$$P = \frac{\text{Dipole Moment}}{\text{Volume}} = \frac{q \times d}{V}$$

Dimensions of Charge (q) = $[AT]$

Dimensions of Distance (d) = $[L]$

Dimensions of Volume (V) = $[L^3]$

$$[P] = \frac{[AT] \times [L]}{[L^3]} = [L^{-2}T^1A^1]$$

(NCERT Ed. 2025-26, 12th, Page No. 78)

2. (2)

The charge on the inner shell be $+Q$.

Due to electrostatic induction, a charge $-Q$ will be induced on the inner surface of the outer shell.

Since, the outer shell is earthed, the potential of the outer shell is zero, but the charge on the inner surface is strictly determined by the enclosed charge.

Charge on inner surface of outer shell = $-Q$.

(NCERT Ed. 2025-26, 12th, Page No. 52)

3. (3)

Original Resistance, $R = \frac{V^2}{P} = \frac{200 \times 200}{100} = 400\,\Omega$.

Cut into two identical parts $\Rightarrow$ Resistance of each part, $r = \frac{R}{2} = 200\,\Omega$.

Connected in parallel, equivalent resistance

$$R_{eq} = \frac{r}{2} = \frac{200}{2} = 100\,\Omega.$$

New Power, $P' = \frac{V^2}{R_{eq}} = \frac{200 \times 200}{100} = 400\,\text{W}$

(NCERT Ed. 2025-26, 12th, Page No. 92)

4. (2)

Using the formula, $\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$

$$\vec{E} = \frac{-17.7 \times 10^{-12}}{8.85 \times 10^{-12}}\hat{i}$$

$$\vec{E} = -2\hat{i}\ \text{N/C}$$

(NCERT Ed. 2025-26, 12th, Page No. 63)

5. (2)

Work done

$$W = pE(\cos\theta_1 - \cos\theta_2) = pE(\cos 60° - \cos 180°)$$

$$= pE(0.5 - (-1)) = 1.5pE = \frac{3}{2}pE$$

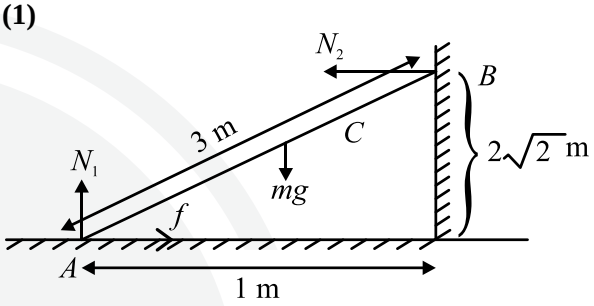
Therefore, $pE = \frac{2W}{3}$

Initial Torque, $\tau = pE\sin 60° = pE\frac{\sqrt{3}}{2}$

$$\tau = \left(\frac{2W}{3}\right)\left(\frac{\sqrt{3}}{2}\right) = \frac{W}{\sqrt{3}}$$

(NCERT Ed. 2025-26, 12th, Page No. 60)

6. (1)



From the FBD of ladder:

For translational equilibrium

$N_1 = mg = 200\,\text{N}$

And $f = N_2$

For rotational equilibrium (About A)

$$mg\left(\frac{1}{2}\right) = N_2(2\sqrt{2})$$

$$\frac{200}{4\sqrt{2}} = N_2$$

$\therefore$ $N_2 = 35.25\,\text{N} \approx 35\,\text{N}$

and $N_1 = 200$ N

(NCERT Ed. 2025-26, 11th, Page No. 114)

7. (4)

$$\varepsilon_{eq} = \frac{\Sigma\left(\frac{\varepsilon_i}{r_i}\right)}{\Sigma\left(\frac{1}{r_i}\right)} = \frac{\frac{2}{1} + \frac{1}{1} - \frac{3}{1}}{\frac{1}{1} + \frac{1}{1} + \frac{1}{1}} = 0$$

(NCERT Ed. 2025-26, 12th, Page No. 97)

8. (3)

Redistribution of charge takes place until potentials are equal. The final electrostatic potential energy of the system is always less than the sum of initial potential energies ($\Delta U = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 - V_2)^2$).

This loss appears as heat/spark. The reason correctly explains the assertion.

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9. (2)

$V = E - Ir$

At $I = 0\,\text{A}, V = 20\,\text{V}$.

$20 = E - (0)r$

$E = 20\,\text{V}$

At $I = 10\,\text{A}, V = 0\,\text{V}$.

$0 = 20 - 10r$

$10r = 20 \Rightarrow r = 2\Omega$

Maximum power is delivered to the external circuit when the external resistance (R) is equal to the internal resistance of the source (r).

$R = r = 2\Omega$

$P_{\max} = \dfrac{E^2}{4r}$

$P_{\max} = \dfrac{(20)^2}{4 \times 2}$

$P_{\max} = \dfrac{400}{8}$

$P_{\max} = 50\,\text{W}$

(NCERT Ed. 2025-26, 12th, Page No. 103)

10. (2)

Distance from vertex to centroid in equilateral triangle $r = \dfrac{L}{\sqrt{3}}$. Potential at centroid

$V = 3 \times \dfrac{kq}{r} = \dfrac{3kq}{L/\sqrt{3}} = \dfrac{3\sqrt{3}kq}{L}$.

Work done, $W = q \times V = \dfrac{3\sqrt{3}kq^2}{L}$.

(NCERT Ed. 2025-26, 12th, Page No. 57)

11. (3)

$P \propto T^4$ and $P' \propto T'^4$

As $P' = 256P \Rightarrow T' = 4T$

$\lambda T = b$ and $\lambda' \times 4T = b$

$\therefore\ \lambda' = \dfrac{\lambda}{4}$

(NCERT Ed. 2025-26, 11th, Page No. 219)

12. (1)

$R_{25} = \dfrac{220^2}{25}$, $R_{100} = \dfrac{220^2}{100}$

$R_{25} = 4R_{100}$

Total $R = 5R_{100}$

Current, $I = \dfrac{440}{5R_{100}}$

Voltage across 25W bulb:

$V_{25} = I \times R_{25} = \dfrac{440}{5R_{100}} \times R_{25} = 440 \times \dfrac{4}{5} = 352\,\text{V}$

Since, $352\,\text{V} > 220\,\text{V}$ (rated), the 25W bulb fuses.

Also $V_{100} = 440 - 352 = 88$ V

Since 88 V < 220 V (rated)

$\therefore$ 100 W bulb will not fuse.

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13. (1)

$V = I_g (G + R)$

$10 = 10 \times 10^{-3} (100 + R)$

$1000 = 100 + R$

$R = 900\Omega$

(NCERT Ed. 2025-26, 12th, Page No. 131)

14. (1)

Force between plates, $F = \dfrac{Q^2}{2A\epsilon_0}$

This formula is independent of distance d. Since Q is constant and A is constant, Force remains F.

(NCERT Ed. 2025-26, 12th, Page No. 74)

15. (3)

$V(r) = \dfrac{kQ}{2R^3}(3R^2 - r^2)$

The potential at the surface ($r = R$) is given by:

$V_{\text{surface}} = \dfrac{kQ}{R} = 80\ \text{V}$

$V_{\text{internal}} = \dfrac{kQ}{2R^3}\left[3R^2 - \left(\dfrac{R}{2}\right)^2\right]$

$V_{\text{internal}} = \dfrac{kQ}{2R^3}\left[3R^2 - \dfrac{R^2}{4}\right]$

$V_{\text{internal}} = \left(\dfrac{kQ}{R}\right) \times \dfrac{11}{8}$

$V_{\text{internal}} = 80 \times \dfrac{11}{8} = 110\ \text{V}$

(NCERT Ed. 2025-26, 12th, Page No. 52)

16. (1)

$E = \dfrac{k|q|}{r^2}$

$|q| = \dfrac{Er^2}{k} = \dfrac{\left(1.5 \times 10^3\right) \times (0.2)^2}{9 \times 10^9}$

$|q| = \dfrac{1.5 \times 0.04}{9} \times 10^{-6} = \dfrac{0.06}{9} \times 10^{-6}$

$= 0.00667 \times 10^{-6}\,\text{C} = 6.67\,\text{nC}$

Since, field is inward, charge is negative. $q = -6.67\,\text{nC}$.

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17. (3)

Consider path *ACDB*,

$V_A + 1 + 2(1) - 2 = V_B$

$V_B = 1 \text{ volt}$

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18. (1)

$V_A - V_B = -\vec{E} \cdot (\vec{r}_A - \vec{r}_B)$

$(\vec{r}_A - \vec{r}_B) = (0-2)\hat{i} + (2-0)\hat{j} = (-2\hat{i} + 2\hat{j})\text{m}$

$\vec{E} \cdot (\vec{r}_A - \vec{r}_B) = (3\hat{i} + 4\hat{j}) \cdot (-2\hat{i} + 2\hat{j})$

$= (3)(-2) + (4)(2) = -6 + 8 = +2$

$\therefore\ V_A - V_B = -(+2) = -2\text{V}$

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19. (2)

Since, the particle moves from distance r to infinity:

Gain in *K.E.* = Initial *P.E.*

$K_f = \frac{kQq}{r}$

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20. (3)

$Q_{AB} = nC_P\Delta T = n\frac{5}{2}R(2T_0 - T_0) = \frac{5}{2}nRT_0$

$Q_{BC} = W_{BC} = -nR.2T_0 \ln 2$

$(W_{BC} = -ve)$ take magnitude only

$\Rightarrow \frac{Q_{AB}}{W_{BC}} = \frac{5}{4\ln 2}$

(NCERT Ed. 2025-26, 11th, Page No. 235)

21. (2)

Drift velocity is given by $v_d = \frac{eE}{m}\tau = \frac{eV}{mL}\tau$.

When the wire is stretched to double its length ($L' = 2L$), the potential V remains constant.

New drift velocity, $v_d' = \frac{eV}{m(2L)}\tau = \frac{1}{2}\left(\frac{eV}{mL}\tau\right) = \frac{v_d}{2}$.

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22. (1)

(1) Formula: $C = \frac{\epsilon_0 A}{d}$. If $d' = 2d$, then

$C' = \frac{\epsilon_0 A}{2d} = \frac{1}{2}C$.

(2) Formula: $E = \frac{V}{d} = \frac{Q}{Cd} = \frac{Q}{\left(\frac{\epsilon_0 A}{d}\right)d} = \frac{Q}{\epsilon_0 A}$.

Since the battery is disconnected, Charge (Q) is constant. Area (A) is constant. Therefore, E does not depend on distance d for an isolated capacitor.

(3) Formula: $V = \frac{Q}{C}$. Since Q is constant and C becomes $\frac{1}{2}$ (from part A), $V' = \frac{Q}{C/2} = 2\frac{Q}{C} = 2V$.

(4) Formula: $U = \frac{1}{2}CV^2$. Since the battery is connected, Potential (V) is constant.

With dielectric $K = 4$, new capacitance $C' = KC = 4C$.

$U' = \frac{1}{2}(4C)V^2 = 4\left(\frac{1}{2}CV^2\right) = 4U$.

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23. (3)

$C = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{K}\right)}$

For Glass, Mica and Ebonite, K is a finite number, so $\left(1 - \frac{1}{K}\right)$ is less than 1.

For Copper (a metal/conductor), the dielectric constant $K \to \infty$.

$C_{\text{copper}} = \frac{\varepsilon_0 A}{d - t(1-0)} = \frac{\varepsilon_0 A}{d - t}$

Since, the denominator ($d - t$) is the smallest possible value compared to the insulators, the capacitance C is maximum for the copper sheet.

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24. (4)

Ohm's law $(V \propto I)$ implies a unique, linear relationship independent of polarity. It fails in all the given cases:

(A) In non-ohmic devices (like semiconductor diodes), the graph of V vs I is a curve, not a straight line.

(B) In devices like diodes, the current differs for the same voltage magnitude depending on direction (forward vs. reverse bias).

(C) In materials like Gallium Arsenide (GaAs), there is a negative resistance region where one value of current corresponds to multiple values of voltage (non-unique).

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25. (3)

Area under velocity *vs* time graph gives displacement and net area under velocity *vs* time graph can be zero here.

Its position *vs* time graph is upward parabola because its acceleration is positive and constant.

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26. (3)

Metal: Charge density is determined by the atomic structure and is already maximum; hence, n remains nearly constant.

Semiconductor & Insulator: Thermal energy breaks covalent bonds, liberating more electrons into the conduction band; hence, n increases.

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27. (3)

Initial Energy, $U_i = \frac{1}{2}CV^2$.

Common potential, $V_c = \frac{Q_{total}}{C_{total}} = \frac{CV+0}{C+2C} = \frac{V}{3}$.

Final Energy,

$U_f = \frac{1}{2}(3C)V_c^2 = \frac{1}{2}(3C)\frac{V^2}{9} = \frac{1}{6}CV^2$

$\Delta U = \frac{1}{2}CV^2 - \frac{1}{6}CV^2 = \frac{3-1}{6}CV^2 = \frac{1}{3}CV^2$.

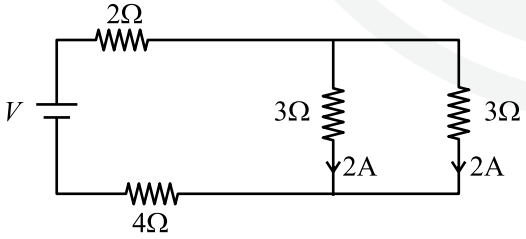
Percentage Loss in energy,

$\frac{\Delta U}{U_i} \times 100 = \frac{1/3}{1/2} \times 100 = \frac{2}{3} \times 100 = 66.67\%$.

(NCERT Ed. 2025-26, 12th, Page No. 76)

28. (3)

Simplified circuit diagram



$i_{net} = 2A + 2A = 4\,A$

$R_{net} = 4 + 4 + \frac{3}{2} = 9.5\Omega$

$V = i_{net}R_{net} = 4 \times 9.5 = 38\,V$

(NCERT Ed. 2025-26, 12th, Page No. 99)

29. (2)

$y = \sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$

$\Rightarrow T = \frac{2\pi}{\omega}$

(NCERT Ed. 2025-26, 11th, Page No. 262)

30. (2)

$V_{R_1} = \frac{\varepsilon R_1}{R_1 + R_2}$

$\Rightarrow V_{30\Omega} = \left(\frac{5}{5+30}\right) \times 30 = \frac{5 \times 30}{35}$

$V_{R_1} = \frac{30}{7}$ volt

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31. (3)

$R_{eq} = R_1 + R_2 = 3R$.

$R_{eq}\left(1 + \alpha_{eq}\Delta T\right) = R\left(1 + \alpha\Delta T\right) + 2R\left(1 + 2\alpha\Delta T\right)$

$3R + 3R\alpha_{eq}\Delta T = 3R + R\alpha\Delta T + 4R\alpha\Delta T$

$3R\alpha_{eq} = 5R\alpha$

$\alpha_{eq} = \frac{5}{3}\alpha$

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32. (4)

$B = \frac{\Delta P}{\frac{-\Delta V}{V}}$

For rigid body, ideal liquid and isochoric process,

$\frac{\Delta V}{V} = 0$

$B = \infty$

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33. (1)

$y = B_0 + A\sin\omega t + B\cos\omega t$

$y - B_0 = \left(\sqrt{A^2 + B^2}\right)\sin\left(\omega t + \phi\right)$

Amplitude $= \sqrt{A^2 + B^2}$

(NCERT Ed. 2025-26, 11th, Page No. 262)

34. (2)

Work done by conservative force depend on initial and final position only. K.E. depend on reference frame.

(NCERT Ed. 2025-26, 11th, Page No. 78)

35. (3)

Since, the total wire length is 200 cm:

$\frac{R}{S} = \frac{l}{200 - l}$

Given $R = 20\,\Omega,\ l = 80$ cm:

$\frac{20}{S} = \frac{80}{200 - 80}$

$\frac{20}{S} = \frac{80}{120}$

$\frac{20}{S} = \frac{2}{3} \Rightarrow S = 30\Omega$

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36. (4)

At the feet, height of the blood column is more as compared to that at brain. Hence, blood pressure in humans is greater at the feet than at brain.

Angle of contact of mercury with glass is obtuse while that of water with glass is acute.

Spinning of cricket ball does not follow a straight trajectory.

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37. (2)

Total length, $L \rightarrow 360° \rightarrow 12\Omega$.

Smaller arc (60°) length is $\frac{1}{6}$ of total.

$R_1 = \frac{12}{6} = 2\Omega$.

$R_2 = 12 - 2 = 10\Omega$.

R_1 and R_2 are in parallel.

$R_{eq} = \frac{2\times10}{2+10} = \frac{20}{12} = \frac{5}{3}\Omega$.

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38. (2)

$v = \sqrt{\frac{2GM}{R}}$

$v' = \sqrt{\frac{2GM}{2R}} = \frac{1}{\sqrt{2}}v$

(NCERT Ed. 2025-26, 11th, Page No. 136)

39. (3)

$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$

$\frac{\partial V}{\partial x} = 6x$

$\frac{\partial V}{\partial y} = -1$

$\frac{\partial V}{\partial z} = 1$

At $(x, y, z) = (1\text{m}, 2\text{m}, 1\text{m})$

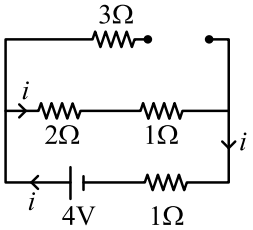
$\vec{E} = -\left(6\hat{i} - 1\hat{j} + 1\hat{k}\right) = \left(-6\hat{i} + \hat{j} - \hat{k}\right)$ N/C

$\therefore \left|\vec{E}\right| = \sqrt{(-6)^2 + (1)^2 + (-1)^2} = \sqrt{36+1+1} = \sqrt{38}$ N/C.

(NCERT Ed. 2025-26, 12th, Page No. 55)

40. (4)

In steady state,



$4 = i \times 4 \Rightarrow i = 1\text{A}$

$V_{\text{across capacitor}} = 3i = 3 \times 1 = 3$ volt

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41. (1)

Beat frequency = 7 – 5 = 2 Hz

Time between two consecutive maxima

$= \frac{1}{f_{Beat}} = \frac{1}{2}$ s

(NCERT Ed. 2025-26, 11th, Page No. 294)

42. (1)

$a_c = \frac{v^2}{r} = \frac{4}{1} = 4\text{ m/s}^2$

(NCERT Ed. 2025-26, 11th, Page No. 41)

43. (3)

$L = m(u\cos\theta)H = 1\times20\times\frac{1}{2}\times400\times\frac{3}{4}\times\frac{1}{20}$

$= 150\text{ kg m}^2\text{/s}$

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44. (2)

Resistance of 6Ω is removed as it is balanced wheatstone bridge

$R_{BE} = \frac{4\times8}{4+8} = \frac{8}{3}\Omega$

$R_{\text{Total}} = \left(\frac{8}{3} + 1.5 + 5.5 + \frac{1}{3}\right)\Omega = 10\Omega$

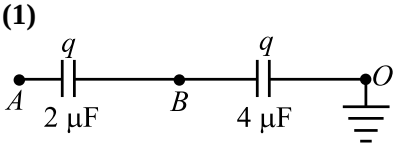
$V = i \times R_{\text{total}}$

$5 = i \times 10$

$i = 0.5\text{A}$

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45. (1)



$V_A - \frac{q}{2} - \frac{q}{4} = 0$

$\Rightarrow 300 = \frac{3q}{4}$

$\Rightarrow q = 400\mu\text{C}$

$\Rightarrow V_B - V_0 = \frac{400\mu\text{C}}{4\mu\text{F}} = 100\text{ V}$

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46. (1)

- Pyrene is used as a fire extinguisher.
- Freons are used in refrigeration and air conditioning.

(NCERT Ed. 2025-26 12th Part-II Page No. 188)

47. (4)

Since unit of k is M s^{-1} it means it is a zero order reaction.

$$t_{completion} = \frac{[A_0]}{k} = \frac{0.1}{0.4} = 0.25\,s$$

(NCERT Ed. 2025-26 12th Part-I Page No. 71)

48. (2)

$$\text{Mole of O-atoms} = \frac{8}{16} = 0.5$$

$$\text{Mole of } CaSO_4 = \frac{1}{4} \times 0.5 = 0.125$$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

49. (2)

The presence of $-NO_2$ group at ortho- and para-positions increases the reactivity of haloarenes towards ArS_N reactions. It is because $-NO_2$ group withdraws the electron density from the benzene ring thus facilitates the attack of the nucleophile on haloarenes also the carbanion formed is stablised through resonance.

(NCERT Ed. 2025-26 12th Part-II Page No. 182)

50. (4)

$$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2OH \xrightarrow[573\ K]{Cu} CH_3-\underset{\underset{CH_3}{|}}{CH}-CHO + H_2$$

(NCERT Ed. 2025-26 12th Part-II Page No. 210)

51. (2)

- Higher the value of (n + *l*), higher will be the energy of orbital
- For the same value of (n + *l*), higher value of n represents higher energy of orbital
- $n = 4, \ell = 1, n + \ell = 5$
- $n = 3, \ell = 2, n + \ell = 5$
- $n = 5, \ell = 0, n + \ell = 5$
- $n = 4, \ell = 0, n + \ell = 4$

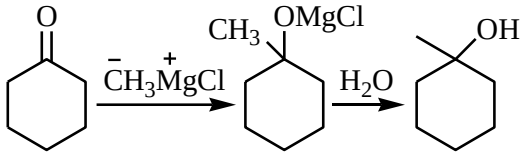
(NCERT Ed. 2025-26 11th Part-I Page No. 61)

52. (2)

A catalyst does not change the Gibb's energy of the reaction.

(NCERT Ed. 2025-26 12th Part-I Page No. 82)

53. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 201)

54. (3)

KCN is the salt of weak acid and strong base. So, their aqueous solutions turns red litmus paper blue.

(NCERT Ed. 2025-26 11th Part-I Page No. 202)

55. (4)

$$k = \frac{2.303}{t}\log\frac{[A_0]}{[A]}$$

$$= \frac{2.303}{4.606}\log\left(\frac{100}{10}\right)$$

$$= 0.5\,s^{-1}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 73)

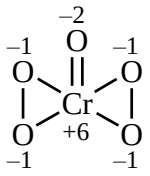
56. (3)

Acidic nature

$$\underset{(-M)}{3} > \underset{\left(\begin{smallmatrix}-M\\ \text{and}\\ \text{H-bonding}\end{smallmatrix}\right)}{2} > \underset{(+I)}{4} > \underset{(H)}{2}$$

(NCERT Ed. 2025-26 12th Part-II Page No. 207)

57. (3)



(NCERT Ed. 2025-26 11th Part-II Page No. 240)

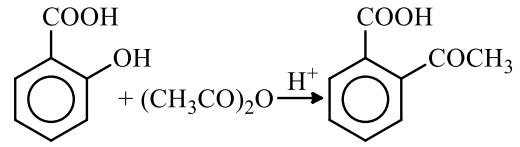
58. (3)

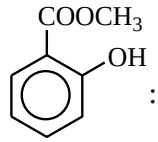
The reaction, $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$, is an example of first order reaction

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[Contd...

59. (1)

- 

Salicyclic acid (Aspirin) + CH_3COOH
- 

: Methyl salicylate.

(NCERT Ed. 2025-26 12th Part-II Page No. 208)

60. (2)

$$\frac{1}{2}X(g) \underset{}{\overset{(K_1)^{1/2}}{\rightleftharpoons}} Y(g)$$

$$Y(g) \overset{K_2}{\rightleftharpoons} 4Z(g)$$

$$\therefore \frac{1}{2}X(g) \overset{(K_1)^{1/2}K_2}{\rightleftharpoons} 4Z(g)$$

$$\therefore 4Z(g) \overset{\frac{1}{\sqrt{K_1}K_2}}{\rightleftharpoons} \frac{1}{2}X(g)$$

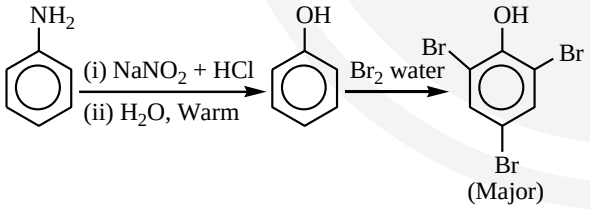
(NCERT Ed. 2025-26 11th Part-I Page No. 176)

61. (2)

For a chemical reaction with rise in temperature by 10°C, the rate constant is nearly doubled

(NCERT Ed. 2025-26 12th Part-I Page No. 78)

62. (1)



(Major)

(NCERT Ed. 2025-26 12th Part-II Page No. 212)

63. (3)

$CuSO_4$ and pyridine are used for denaturation of commercial alcohol.

(NCERT Ed. 2025-26 12th Part-II Page No. 214)

64. (1)

1-Chloropropane does not contain any chiral centre.

(NCERT Ed. 2025-26 12th Part-II Page No. 176)

65. (3)

Ortho and para nitrophenol can be separated by steam distillation because o-nitrophenol is steam volatile due to intramolecular H-bonding while p-nitrophenol is less volatile due to intermolecular H-bonding.

(NCERT Ed. 2025-26 12th Part-II Page No. 211)

66. (2)

ΔU does not discriminate between reversible and irreversible process

(NCERT Ed. 2025-26 11th Part-I Page No. 159)

67. (4)

- Diastereoisomers may be optically inactive.
- Racemic mixture is always optically inactive.

(NCERT Ed. 2025-26 12th Part-II Page No. 177)

68. (1)

Phenoxide ion is more reactive than phenol.

(NCERT Ed. 2025-26 12th Part-II Page No. 212)

69. (2)

IUPAC group prefix used for ethers is alkoxy.

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

70. (2)

Molecularity is applicable only for elementary reactions

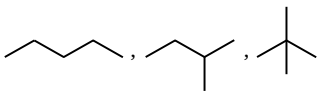
(NCERT Ed. 2025-26 12th Part-I Page No. 70)

71. (2)

The C–O–C, bond angle in CH_3OCH_3 is slightly greater than the tetrahedral angle due to repulsive interaction between the two methyl groups.

(NCERT Ed. 2025-26 12th Part-II Page No. 199)

72. (2)



(NCERT Ed. 2025-26 11th Part-II Page No. 270)

73. (1)

1,4-Dichlorobenzene has more melting point than 1,2-dichlorobenzene.

(NCERT Ed. 2025-26 12th Part-II Page No. 169)

74. (2)

Williamson synthesis:

$$\underset{(1^\circ)}{R-X} + R'-O^{\ominus} \xrightarrow{S_N2} R-O-R' + X^-$$

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[Contd...

75. (2)

For an ideal black body. Intensity of radiation depends only on its temperature.

(NCERT Ed. 2025-26 11th Part-I Page No. 40)

76. (1)

E_2 elimination take place by Sytzeff rule and more stable alkene is the major product.

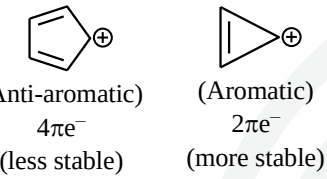
(NCERT Ed. 2025-26 11th Part-II Page No. 310)

77. (1)

$$CH_3-O-CH_2CH_3 \xrightarrow[S_N2]{\text{Conc. HI}} CH_3I + CH_3CH_2OH$$

(NCERT Ed. 2025-26 12th Part-II Page No. 218)

78. (2)



(Anti-aromatic) $4\pi e^-$ (less stable) (Aromatic) $2\pi e^-$ (more stable)

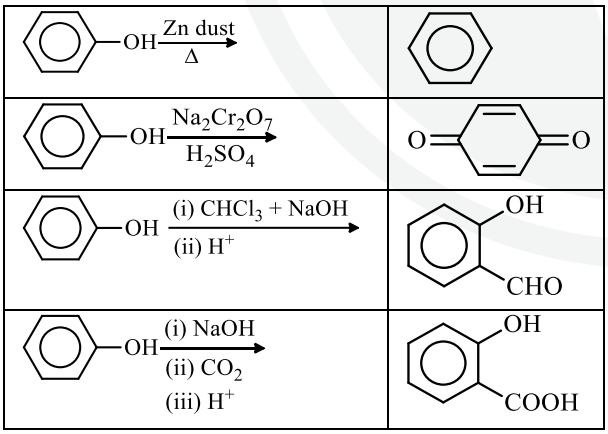
(NCERT Ed. 2025-26 11th Part-II Page No. 272)

79. (3)

B.P: RI > RBr > RCl > RF

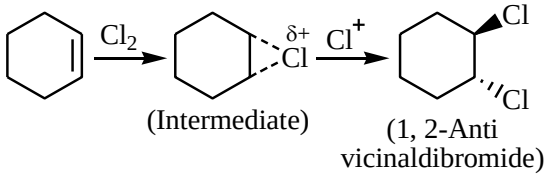
(NCERT Ed. 2025-26 12th Part-II Page No. 168)

80. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 213)

81. (2)

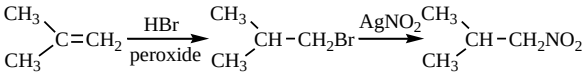


(Intermediate) (1, 2-Anti vicinaldibromide)

Cyclic halonium ion is formed as an intermediate in the reaction.

(NCERT Ed. 2025-26 11th Part-II Page No. 311)

82. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 170)

83. (2)

A better reagent for oxidation of primary alcohols to aldehydes in good yield is PCC.

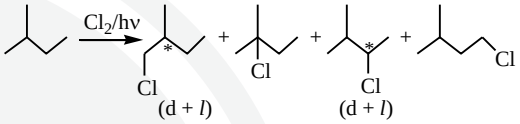
(NCERT Ed. 2025-26 12th Part-II Page No. 210)

84. (3)

Boiling point of n-butane is more than that of propane because there is more van der Waals forces among n-butane molecules due to more surface area than propane.

(NCERT Ed. 2025-26 11th Part-II Page No. 302)

85. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 177)

86. (1)

Nowadays, large quantity of ethanol are obtained by hydration of ethene

(NCERT Ed. 2025-26 12th Part-II Page No. 214)

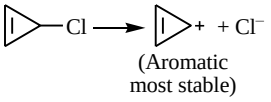
87. (3)

Vapour pressure is not a colligative property.

(NCERT Ed. 2025-26 12th Part-I Page No. 214)

88. (2)

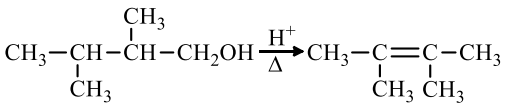
More the stability of intermediate carbocation, more will the reactivity of alkyl halide towards S_N1 reaction



(NCERT Ed. 2025-26 12th Part-II Page No. 173)

89. (2)

Rearrangement take place during dehydration



(NCERT Ed. 2025-26 12th Part-II Page No. 208)

90. (2)

$$\underset{1\,mol}{\overset{0}{I_2}} \rightarrow 2I^{+7} + \underset{14\,F}{14e^-}$$

Required faraday = 14 × 0.5 = 7 F

(NCERT Ed. 2025-26 12th Part-I Page No. 51)

[Contd...

91. (1)

In the Hershey-Chase experiment , viruses grown in the presence of radioactive phosphorus contained radioactive DNA but not radioactive protein because DNA contains phosphorus but protein does not. The experiment concluded that DNA is the genetic material that is passed from virus to bacteria.

(NCERT Ed. 2025-26, 12th, Page No. 86)

92. (2)

Chrysophytes include diatoms and golden algae (desmids). In diatoms the cell walls form two thin overlapping shells. The walls are embedded with silica and thus the walls are indestructible. Thus, diatoms have left behind large amount of cell wall deposits in their habitat; this accumulation over billions of years is referred to as 'diatomaceous earth'. Diatoms are the chief 'producers' in the oceans. Red tides are primarily caused by red dinoflagellates. All protozoans are heterotrophs and live as predators or parasites.

(NCERT Ed. 2025-26, 11th, Page No. 12)

93. (2)

In flowering plants, most zygotes divide only after certain amount of endosperm is formed. This is an adaptation to provide assured nutrition to the developing embryo.

(NCERT Ed. 2025-26, 12th, Page No. 18)

94. (4)

In a dicot root, all tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith constitute the stele. Monocotyledonous roots do not undergo any secondary growth. Pith is large and well developed in monocot root. There are usually more than six (polyarch) xylem bundles in the monocot root as compared to the dicot root which have fewer xylem bundles.

(NCERT Ed. 2025-26, 11th, Page No. 74)

95. (3)

E. coli divides in 20 minutes.

(NCERT Ed. 2025-26, 12th, Page No. 89)

96. (4)

Selaginella is a pteridophyte. It has motile, biflagellate male gametes (antherozoids) that need water to reach the female gamete. However, the female gamete, or egg, is non-motile and is retained within the archegonium of the female gametophyte.

(NCERT Ed. 2025-26, 11th, Page No. 32)

97. (1)

Following double fertilisation, events of endosperm and embryo development, maturation of ovule(s) into seed(s) and ovary into fruit, are collectively termed post-fertilisation events. The central cell of embryo sac after triple fusion becomes the primary endosperm cell (PEC) and develops into the endosperm while the zygote develops into an embryo. The synergids and antipodals degenerate and integuments form the seed coat.

(NCERT Ed. 2025-26, 12th, Page No. 18)

98. (2)

In prokaryotes, the RNA polymerase is only capable of catalysing the process of elongation. It associates transiently with initiation-factor (σ) and termination-factor (ρ) to initiate and terminate the transcription, respectively. Association with these factors alter the specificity of the RNA polymerase to either initiate or terminate. If the ρ factor is not produced due to a mutation, the process of Rho-dependent termination will fail.

(NCERT Ed. 2025-26, 12th, Page No. 94)

99. (1)

During leptotene stage the chromosomes become gradually visible under the light microscope. During zygotene, chromosomes start pairing together and this process of association is called synapsis. The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad. In oocytes of some vertebrates, diplotene can last for months or years. The final stage of meiotic prophase I is diakinesis. During this phase the chromosomes are fully condensed and the meiotic spindle is assembled to prepare the homologous chromosomes for separation.

(NCERT Ed. 2025-26, 11th, Page No. 126)

100. (3)

The MMC undergoes meiotic division. Meiosis results in the production of four megaspores. Only one of the megaspores is functional while the other three degenerate. So, 1 MMC - 1 meiotic division - 1 functional megaspore. For 100 embryo sacs, 100 meiotic divisions are needed. The nucleus of the functional megaspore divides mitotically to form two nuclei. Two more sequential mitotic nuclear divisions result in the formation of the 4-nucleate and later the 8-nucleate stages. This means 1 functional megaspore undergoes 3 mitotic divisions to form one mature embryo sac. For 100 embryo sacs, $100 \times 3 = 300$ mitotic divisions are needed.

(NCERT Ed. 2025-26, 12th, Page No. 10)

[Contd...

101. (1)
The DNA profile of B does not match the crime scene sample (A), so it cannot be the criminal. The banding pattern of DNA fingerprinting is exactly same in the case of monozygotic (identical) twins. Therefore, while it is true that we cannot distinguish between C and D based on this DNA test alone, it is possible that both the twins were involved in the crime. The DNA evidence simply places their genetic profile at the scene; it does not rule out the participation of both. As seen in the gel, the specific polymorphic regions (the bands) are identical across samples A, C, and D.
(NCERT Ed. 2025-26, 12th, Page No. 106)

102. (2)
Kingdom (C. Fungi)
Class (A. Pteropsida)
Order (E. Primata)
Family (B. Felidae)
Genus (D. *Brasssica*)
The correct sequence is therefore C → A → E → B → D.
(NCERT Ed. 2025-26, 11th, Page No. 7)

103. (2)
Although in most of the species, fruits are the results of fertilisation, there are a few species in which fruits develop without fertilisation. Such fruits are called parthenocarpic fruits. Banana is one such example. Parthenocarpy can be induced through the application of growth hormones and such fruits are seedless.
(NCERT Ed. 2025-26, 12th, Page No. 21)

104. (2)
The human genome contains 3164.7 million bp. The total number of genes is estimated at 30,000. Less than 2 per cent of the genome codes for proteins. Almost all (99.9 per cent) nucleotide bases are exactly the same in all humans.
(NCERT Ed. 2025-26, 12th, Page No. 104)

105. (1)
In the hypogynous flower the gynoecium occupies the highest position while the other parts are situated below it. The ovary in such flowers is said to be superior.
(NCERT Ed. 2025-26, 11th, Page No. 63)

106. (4)
Majority of plants use biotic agents for pollination. Insects are the most dominant biotic agents of pollination. Among the animals, insects, particularly bees are the dominant biotic pollinating agents.
(NCERT Ed. 2025-26, 12th, Page No. 12)

107. (3)
DNA chemically is less reactive and structurally more stable when compared to RNA. In fact, the presence of thymine at the place of uracil also confers additional stability to DNA.
(NCERT Ed. 2025-26, 12th, Page No. 87)

108. (1)
In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. Nucleus as a cell organelle was first described by Robert Brown as early as 1831. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells. In 1838, Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant. At about the same time, Theodore Schwann (1839), a German Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape. Cell theory as understood today is:
(i) all living organisms are composed of cells and products of cells.
(ii) all cells arise from pre-existing cells.
(NCERT Ed. 2025-26, 11th, Page No. 88)

109. (3)
The codon is triplet. 61 codons code for amino acids and 3 codons do not code for any amino acids, hence they function as stop codons. The codon is read in mRNA in a contiguous fashion. There are no punctuations. The code is nearly universal. Some exceptions to this rule have been found in mitochondrial codons, and in some protozoans.
(NCERT Ed. 2025-26, 12th, Page No. 96)

110. (2)
The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte.
(NCERT Ed. 2025-26, 11th, Page No. 29)

111. (2)
The hard outer layer called the exine is made up of sporopollenin which is one of the most resistant organic material known. It can withstand high temperatures and strong acids and alkali. No enzyme that degrades sporopollenin is so far known.

[Contd...

Dehydration and dormancy of mature seeds are crucial for storage of seeds which can be used as food throughout the year and also to raise crop in the next season.

(NCERT Ed. 2025-26, 12th, Page No. 7)

112. (4)

A typical nucleosome contains 200 bp of DNA helix.

(NCERT Ed. 2025-26, 12th, Page No. 83)

113. (1)

Dorsiventral (Dicotyledonous) Leaf : The tissue between the upper and the lower epidermis is called the mesophyll. Mesophyll, which possesses chloroplasts and carry out photosynthesis, is made up of parenchyma. The epidermis which covers both the upper surface (adaxial epidermis) and lower surface (abaxial epidermis) of the leaf has a conspicuous cuticle. The abaxial epidermis generally bears more stomata than the adaxial epidermis. The veins vary in thickness in the reticulate venation of the dicot leaves.

(NCERT Ed. 2025-26, 11th, Page No. 76)

114. (3)

The antipodal cell, microspore, and egg cell are all haploid (n) structures. The megaspore mother cell is the only diploid (2n) structure in the list, making it the odd one out.

(NCERT Ed. 2025-26, 12th, Page No. 10)

115. (1)

QB bacteriophage has RNA as the genetic material. *Caenorhabditis elegans* is a free living non-pathogenic nematode.

(NCERT Ed. 2025-26, 12th, Page No. 87)

116. (4)

Solanum tuberosum is potato. *Solanum* includes species like *nigrum* and *melongena*. *aestivum* is a species in genus *Triticum*.

(NCERT Ed. 2025-26, 11th, Page No. 7)

117. (4)

The plant shown in the question is one that produces both closed and open flowers. These flowers are bisexual. Autogamy can occur in both the closed and open type of flowers. Geitonogamy can occur in its open flowers but it does not bring about genetic variations in its progeny. Xenogamy is possible in open flowers (A) but not in closed flowers (B).

(NCERT Ed. 2025-26, 12th, Page No. 12)

118. (2)

In eukaryotes, the primary transcripts contain both the exons and the introns and are non-functional. Hence, it is subjected to a process called splicing where the introns are removed and exons are joined in a defined order. hnRNA undergoes additional processing called as capping and tailing. In capping an unusual nucleotide (methyl guanosine triphosphate) is added to the 5'-end of hnRNA. In tailing, adenylate residues (200-300) are added at 3'-end.

(NCERT Ed. 2025-26, 12th, Page No. 95)

119. (2)

Metaphase is the stage at which morphology of chromosomes is most easily studied.

(NCERT Ed. 2025-26, 11th, Page No. 123)

120. (1)

Wind pollination requires that the pollen grains are light and non-sticky so that they can be transported in wind currents. They often possess well-exposed stamens (so that the pollens are easily dispersed into wind currents) and large often-feathery stigma to easily trap air-borne pollen grains. A familiar example is the corn cob – the ears you see are nothing but the stigma and style which wave in the wind to trap pollen grains.

(NCERT Ed. 2025-26, 12th, Page No. 13)

121. (1)

It was only in 1953 that James Watson and Francis Crick proposed a very simple but famous Double Helix model for the structure of DNA. Frederick Griffith observed transformation in pneumococcus. Prior to the work of Oswald Avery, Colin MacLeod and Maclyn McCarty (1933-44), the genetic material was thought to be a protein. They worked to determine the biochemical nature of 'transforming principle' in Griffith's experiment. Frederick Sanger developed methodology for sequencing of DNA.

(NCERT Ed. 2025-26, 12th, Page No. 84)

122. (3)

The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.

(NCERT Ed. 2025-26, 11th, Page No. 95)

123. (4)

Some examples of water pollinated plants are *Vallisneria* and *Hydrilla* which grow in fresh water and several marine sea-grasses such as *Zostera*. Not all aquatic plants use water for pollination. In a majority of aquatic plants such as water hyacinth and water lily, the flowers emerge above the level of water and are pollinated by insects or wind as in most of the land plants.

(NCERT Ed. 2025-26, 12th, Page No. 13)

124. (3)

The process of translation requires transfer of genetic information from a polymer of nucleotides to synthesise a polymer of amino acids.

(NCERT Ed. 2025-26, 12th, Page No. 95)

125. (4)

When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be axile, as in china rose, tomato and lemon.

(NCERT Ed. 2025-26, 11th, Page No. 65)

126. (1)

In some species floral rewards are in providing safe places to lay eggs; an example is that of the tallest flower of *Amorphophallus* (the flower itself is about 6 feet in height).

(NCERT Ed. 2025-26, 12th, Page No. 14)

127. (3)

The sensitivity of the DNA fingerprinting technique has been increased by use of polymerase chain reaction (PCR).

(NCERT Ed. 2025-26, 12th, Page No. 106)

128. (1)

The deuteromycetes reproduce only by asexual spores known as conidia. *Aspergillus* is a member of Ascomycetes. In ascomycetes, sexual spores are called ascospores which are produced endogenously Yeast is unicellular fungus belonging to Ascomycetes.

(NCERT Ed. 2025-26, 11th, Page No. 17)

129. (2)

Sexual mode of reproduction enables creation of new variants, so that survival advantage is enhanced.

(NCERT Ed. 2025-26, 12th, Page No. 1)

130. (4)

Adenine is a purine, common to both DNA and RNA. Two nucleotides are linked by a phosphodiester bond.

(NCERT Ed. 2025-26, 12th, Page No. 80)

131. (2)

The ovule is a small structure attached to the placenta by means of a stalk called funicle.

(NCERT Ed. 2025-26, 12th, Page No. 9)

132. (4)

In a typical nucleus, some region of chromatin are loosely packed (and stains light) and are referred to as euchromatin. The chromatin that is more densely packed and stains dark are called as Heterochromatin. Euchromatin is said to be transcriptionally active chromatin, whereas heterochromatin is inactive.

(NCERT Ed. 2025-26, 12th, Page No. 84)

133. (1)

In most plants, by the time the fruit develops from the ovary, other floral parts degenerate and fall off. However, in a few species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called false fruits. Most fruits however develop only from the ovary and are called true fruits e.g. mango. Although seeds, in general are the products of fertilisation, a few flowering plants such as some species of *Asteraceae* and grasses, have evolved a special mechanism, to produce seeds without fertilisation, called apomixis. Occurrence of more than one embryo in a seed is referred to as polyembryony e.g. orange.

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134. (3)

A transcription unit in DNA is defined primarily by the three regions in the DNA: (i) A Promoter (ii) The Structural gene (iii) A Terminator. The promoter is said to be located towards 5′ -end (upstream) and terminator towards the 3'-end (downstream) of the structural gene.

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135. (2)

There are several records of very old yet viable seeds. The oldest is that of a lupine, *Lupinus arcticus* excavated from Arctic Tundra. The seed germinated and flowered after an estimated record of 10,000 years of dormancy. A recent record of 2000 years old viable seed is of the date palm, *Phoenix dactylifera* discovered during the archeological excavation at King Herod's palace near the Dead Sea.

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[Contd...

136. (1)

The primary response is slow and low intensity because the body takes time to identify the pathogen and produce antibodies. However, memory cells are formed. Upon subsequent exposure to the same pathogen, these memory cells trigger a rapid and massive production of antibodies (anamnestic response).

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137. (3)

Low K_m means higher affinity of enzyme for substrate. Zinc is a cofactor of carboxypeptidase.

Enzymes lower activation energy, making the substrate → product transition easier.

Conversion of glucose to pyruvic acid (glycolysis) involves 10 enzymatic reactions.

(NCERT Ed. 2025-26, 11th, Page No. 114, 115, 118)

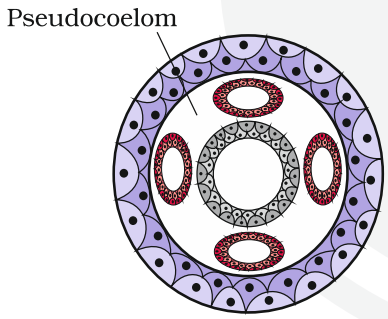
138. (3)

ZIFT involves transfer of a zygote or early embryo (up to 8 blastomeres) into the fallopian tube.

IUI (Intra-Uterine Insemination) involves transfer of semen, not embryos, into the uterus. Embryo transfer (>8 blastomeres) is done in IUT.

(NCERT Ed. 2025-26, 12th, Page No. 48)

139. (3)



In some animals, the body cavity is not lined by mesoderm, instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm. Such a body cavity is called pseudocoelom and the animals possessing them are called pseudocoelomates, e.g., aschelminthes.

(NCERT Ed. 2025-26, 11th, Page No. 39)

140. (2)

List-I	List-II
PCT	Reabsorption of 70-80% electrolytes
Descending Limb of Loop of Henle	Permeable to water, impermeable to electrolytes
DCT	Conditional reabsorption of Na^+ and water
Ascending Limb of Loop of Henle	Impermeable to water, permeable to electrolytes

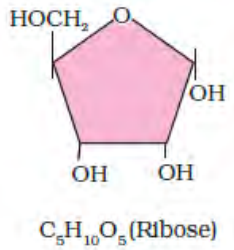
(NCERT Ed. 2025-26, 11th, Page No. 209)

141. (2)

Contact inhibition is the property of normal cells by which they stop dividing when they come in contact with neighboring cells. Loss of this property is characteristic of cancer cells.

(NCERT Ed. 2025-26, 12th, Page No. 141)

142. (3)



$C_5H_{10}O_5$ (Ribose)

Ribose in present in RNA.

(NCERT Ed. 2025-26, 11th, Page No. 107, 111)

143. (2)

Copper-releasing IUDs (e.g., Cu-T, Cu-7, Multiload 375) work by releasing copper ions, which suppress sperm motility and fertilising capacity and increase phagocytosis of sperms in the uterus.

(NCERT Ed. 2025-26, 12th, Page No. 44)

144. (2)

The limbic system is responsible for emotions, motivation, and sexual behaviour.

Voluntary movements are controlled by the cerebral cortex, so they remain unaffected.

Respiration and cardiovascular reflexes are controlled by the medulla oblongata and pons, so they remain intact.

(NCERT Ed. 2025-26, 11th, Page No. 236)

145. (1)

Spermatogenesis begins at puberty due to a surge in GnRH, which stimulates FSH and LH secretion, activating the testes.

Oogenesis starts during fetal development; millions of oogonia are formed in the ovary, most of which degenerate before birth.

(NCERT Ed. 2025-26, 12th, Page No. 31, 32)

146. (2)

Chronological life cycle of *Plasmodium*:

– Sporozoites released from mosquito salivary glands → injected into human during mosquito bite.

– Parasite reproduces asexually in liver cells → forms merozoites.

– Rupture of RBCs and release of haemozoin → causes malaria symptoms.

– Sexual stages (gametocytes) develop in human RBCs → some merozoites differentiate into gametocytes.

– Female mosquito takes up gametocytes with blood meal → continues the cycle in mosquito.

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[Contd...

147. (4)

Second messengers transmit signals from membrane-bound receptors to intracellular targets. Examples: cAMP, IP_3, Ca^{2+}.

Trypsin is a digestive enzyme, not involved in intracellular signaling.

(NCERT Ed. 2025-26, 11th, Page No. 248)

148. (4)

Vinblastin and Curcumin are classified as drugs. Concanavalin A is an example of a lectin.

(NCERT Ed. 2025-26, 11th, Page No. 108)

149. (2)

Abortions can occur spontaneously (miscarriage) or be induced.

Infertility is the inability to produce a viable offspring and can be due to male, female, or combined factors, not only female abnormalities.

Lactational Amenorrhea Method (LAM) can act as a natural contraceptive when breastfeeding is exclusive.

Sex education and awareness improve reproductive health by promoting safe practices and early detection of problems.

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150. (3)

Normally, arterial blood carries ~20 ml O_2 per 100 ml. At rest, tissues extract about 25%, so ~5 ml O_2 is delivered. Binding of O_2 with hemoglobin depends primarily on partial pressure of O_2, and partial pressure of CO_2 does not interfere.

O_2 binding does depend on pO_2 but pCO_2 does influence O_2 binding via the Bohr effect: high CO_2 lowers hemoglobin's O_2 affinity, enhancing O_2 delivery.

(NCERT Ed. 2025-26, 11th, Page No. 189)

151. (3)

Lecithin is a phospholipid found in cell membranes. Palmitic and Arachidonic are fatty acids. Glycerol is a simple lipid component.

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152. (3)

Retroviruses (e.g., HIV) have RNA genomes.

During replication, the viral RNA is converted into complementary DNA (cDNA) by reverse transcriptase.

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153. (2)

Duration of one cardiac cycle is about 0.8 seconds at normal heart rate. Stroke volume ≈ 70 mL per beat in a healthy adult. The lub sound (S_1) is due to closure of atrioventricular (AV) valves, not semilunar valves (which produce the dub sound, S_2). P-wave of ECG represents atrial depolarization.

(NCERT Ed. 2025-26, 11th, Page No. 200, 201)

154. (2)

The statutory ban on amniocentesis in India is primarily to prevent misuse of the technique for sex determination, which led to female foeticide.

Although amniocentesis can detect genetic abnormalities, its legal restriction aims to curb sex-selective abortions.

(NCERT Ed. 2025-26, 12th, Page No. 42)

155. (2)

During contraction, the I-band shortens, the H-zone disappears/shortens, and the Sarcomere shortens. However, the A-band (length of thick myosin filaments) remains constant.

(NCERT Ed. 2025-26, 11th, Page No. 222)

156. (2)

The primary structure of a protein refers to the specific sequence and positional order of amino acids linked by peptide bonds. This sequence determines higher levels of structure.

The α-helix, a common secondary structure, is formed when the polypeptide chain coils in a right-handed manner, stabilized by hydrogen bonds. A left-handed α-helix is not found in proteins under normal biological conditions.

The tertiary structure represents the overall three-dimensional folding of a protein, which is essential for its biological activity.

Haemoglobin consists of four polypeptide chains; hence it shows a quaternary structure.

(NCERT Ed. 2025-26, 11th, Page No. 112)

157. (1)

Heroin (an opioid) binds to specific opioid receptors present in the central nervous system and gastrointestinal tract. It acts as a depressant, slowing down body functions such as respiration and heart rate.

(NCERT Ed. 2025-26, 12th, Page No. 142)

158. (2)

Protonephridia or flame cells are the excretory structures in Platyhelminthes (Flatworms, e.g., *Planaria*), rotifers, some annelids and the cephalochordate – *Amphioxus*.

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[Contd...

159. (2)

Saheli is a non-steroidal, once-a-week oral contraceptive pill developed in India by CDRI, Lucknow. It is reversible and has very few side effects.

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160. (3)

Cellulose is a homopolymer of β-D-glucose units.

Cellulose contains only glucose; fructose is not present. The β-1,4 glycosidic bonds are between glucose units, not glucose and fructose.

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161. (2)

Widal test is the diagnostic test used to confirm typhoid fever by detecting antibodies against *Salmonella typhi*.

Mary Mallon (also known as Typhoid Mary) was a famous asymptomatic carrier of typhoid fever.

(NCERT Ed. 2025-26, 12th, Page No. 131)

162. (1)

Chordates: The central nervous system (CNS) is dorsal, hollow, and single (forms the brain and spinal cord).

Non-chordates: The CNS is ventral, solid, and usually paired or ganglionated (e.g., in annelids and arthropods).

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163. (3)

Genital herpes is caused by Herpes simplex virus (HSV), a viral infection.

Viral STIs like herpes and HIV cannot be completely cured; only symptoms can be managed.

Syphilis, Gonorrhoea, and Chlamydiasis are bacterial infections and can be completely cured with antibiotics if detected early.

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164. (1)

List-I	List-II
Monosaccharide	$C_6H_{12}O_6$
Proteins	Peptide bond
Fatty acid	$CH_3–(CH_2)_{14}–COOH$
Nucleic Acids	Heteropolymer of nucleotides

(NCERT Ed. 2025-26, 11th, Page No. 106, 109, 111)

165. (2)

The parathyroid glands secrete parathyroid hormone (PTH), which plays a critical role in maintaining calcium homeostasis in the body. It is considered a hypercalcemic hormone because it acts to increase Ca^{2+} levels in the blood.

(NCERT Ed. 2025-26, 11th, Page No. 243)

166. (4)

Colostrum, the yellowish fluid secreted by a mother during the initial days of lactation, is exceptionally rich in antibodies that provide essential passive immunity to the newborn.

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167. (2)

One major factor for population explosion is the rapid decline in death rate due to improved healthcare, better nutrition, and sanitation, which allows more people to live longer. Another contributing factor is the decline in maternal mortality rate; with better maternal care, more women survive childbirth, resulting in more children being added to the population. Similarly, the rapid decline in infant mortality rate ensures that more infants survive past early childhood, further increasing population growth.

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168. (2)

Renin is an enzyme secreted by the juxtaglomerular cells of the kidney. It acts on angiotensinogen (produced by the liver) and converts it into angiotensin I. Angiotensin I is then converted into angiotensin II by angiotensin-converting enzyme (ACE) in the lungs.

Angiotensin II ultimately stimulates aldosterone release from the adrenal cortex and causes vasoconstriction, helping regulate blood pressure.

(NCERT Ed. 2025-26, 11th, Page No. 212)

169. (2)

Essential amino acids: Body cannot make them → must come from diet.

Non-essential amino acids: Body can synthesize them → dietary intake not mandatory.

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170. (3)

Dengue is caused by Dengue virus and is transmitted by female *Aedes* mosquito.

Malaria is transmitted by female *Anopheles* mosquito.

Filariasis is transmitted by *Culex* mosquito.

Amoebiasis is caused by *Entamoeba histolytica* and is not vector-borne; it spreads through contaminated food and water.

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[Contd...

171. (1)

List-I	List-II
Fibrous Joint	Skull sutures
Cartilaginous Joint	Between adjacent vertebrae
Pivot Joint	Between Atlas and Axis
Ball and Socket	Between Humerus and Pectoral Girdle

(NCERT Ed. 2025-26, 11th, Page No. 227)

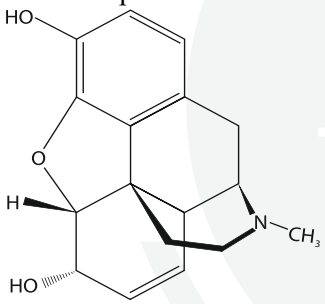
172. (3)

Surgical methods like vasectomy and tubectomy prevent gamete release or fertilization.
Not all STDs are bacterial; viral STDs like HIV and Herpes cannot be cured.
Oral pills are widely used and reversible contraceptives, including in rural areas.
In Embryo Transfer (ET) / IVF, embryos are always transferred into the uterus.

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173. (4)

Heroin commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine.



Chemical structure of Morphine

(NCERT Ed. 2025-26, 12th, Page No. 142)

174. (2)

Allergies involve IgE antibodies binding to mast cells, causing them to release histamine and serotonin, leading to inflammation.

(NCERT Ed. 2025-26, 12th, Page No. 137)

175. (1)

List-I	List-II
Tidal Volume	500 mL
IRV	2500 - 3000 mL
ERV	1000 - 1100 mL
Residual Volume	1100 - 1200 mL

(NCERT Ed. 2025-26, 11th, Page No. 186, 187)

176. (2)

Diaphragms are reusable barrier devices inserted into the vagina to cover the cervix and prevent sperm from entering the uterus.
Female condoms are single-use and disposable.
Male condoms are also single-use.
Implants are hormonal contraceptives.

(NCERT Ed. 2025-26, 12th, Page No. 44)

177. (1)

ABO blood grouping depends on the presence or absence of antigens (A or B) on RBCs.
Plasma contains antibodies against the antigens not present on the RBCs (e.g., group A has anti-B antibodies).
Erythroblastosis fetalis occurs when the mother is Rh− and the fetus is Rh+, not the other way around.
O− blood group is the universal donor, as it has no A, B, or Rh antigens.

(NCERT Ed. 2025-26, 11th, Page No. 195, 196)

178. (4)

In competitive inhibition, the inhibitor competes with the substrate for the active site of the enzyme because it closely resembles the substrate.

(NCERT Ed. 2025-26, 11th, Page No. 117)

179. (3)

Mucosa-associated lymphoid tissue (MALT) constitutes about 50% of the lymphoid tissue in the human body.
MALT is located mainly in the lining of major body tracts such as the respiratory, digestive, and urogenital tracts, which are constantly exposed to pathogens.

(NCERT Ed. 2025-26, 12th, Page No. 138)

180. (2)

At rest, the neuronal membrane is more permeable to K^+ ions due to potassium leak channels. During depolarisation, voltage-gated Na^+ channels open, making the membrane highly permeable to Na^+ ions.

(NCERT Ed. 2025-26, 11th, Page No. 232, 233)

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